Detecting Coronary Heart Disease from Lifestyle Survey Data with NumPy-only Machine Learning

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Abstract

We predict whether a person has coronary heart disease (MICHD) from 321 health and lifestyle answers of the 2015 Behavioral Risk Factor Surveillance System (BRFSS) [1]. Only 8.8 % of the respondents are positive, so we optimize the F1 score and tune the decision threshold instead of using 0.5. We clean the survey answer codes, add one-hot encodings and a few interaction features, and compare the six required linear methods with decision trees, random forests, gradient boosting, an ensemble and a neural network. Our final model, a regularized logistic regression, reaches a 5-fold cross-validated F1 of 0.427 ± 0.004 and an F1 of 0.441 on the AIcrowd test set.

I. Introduction

The training set contains $N = 328{,}135$ respondents described by $D = 321$ survey features; the test set contains 109,379 respondents. The label is $+1$ when the person reported having been told they had a heart attack or angina. Only 28,975 respondents (8.8 %) are positive. A model that always predicts "healthy" therefore reaches 91.2 % accuracy while detecting no sick person, which is why we evaluate models with the F1 score of the positive class, the harmonic mean of precision and recall.

All our models output a score (a probability for the logistic models), and a person is predicted sick when the score is above a threshold t. Because positives are rare, the F1-optimal threshold is far below 0.5 ($t \approx 0.21$ for our final model), so the threshold is always tuned on validation data.

II. Data and preprocessing

Cleaning. BRFSS answers contain special codes: 7/77/777 ("don't know"), 9/99/999 ("refused"), and 88 ("none"). We grouped the columns automatically by their range and code family, then recoded the codes to missing values (NaN), 88 to 0 for counts (e.g. days of poor health), yes/no answers from 1/2 to 1/0, and the categories of diabetes and high blood pressure to $\{0, 0.5, 1\}$ (no / borderline / yes). We removed the columns that are more than 90 % missing or constant, the administrative and phone variables, the survey weights, and raw answers that have a cleaner computed version (e.g. weight in pounds or kilograms). This leaves 170 columns.

Feature engineering. Seven nominal variables (marital status, employment, housing, personal doctor, race, sexual orientation, transgender) are one-hot encoded instead of being used as ordered numbers. We add six products of features suggested by the medical background (age $\times$ high blood pressure, diabetes, general health, sex and stroke; general health $\times$ difficulty walking) and the squares of age and BMI. Finally, 4 columns that are exact copies of others are removed, giving 203 features. Missing values are replaced by the training mean, every feature is standardized with the training statistics, and a bias column is added.

Evaluation protocol. Every rule and statistic is learned on the training part only. Feature and model choices are made on a fixed random 80/20 split (seed 0), with the F1-optimal threshold chosen on the validation part. The final model is assessed with stratified 5-fold cross-validation (CV), the threshold being tuned on an inner 20 % split of each training fold, so the held-out fold is never used for any choice.

III. Models

Six required methods. Linear regression with gradient descent (GD) and stochastic gradient descent (SGD, one sample per step), least squares and ridge regression (normal equations) are trained on the labels $y \in \{0, 1\}$, and their outputs are used as scores. Logistic regression and regularized logistic regression (penalty $\lambda\|w\|^2$) are trained with GD. Step sizes: $\gamma = 1/L$ for MSE-GD, where $L \approx 15$ is the largest eigenvalue of $X^\top X/N$; $\gamma = 10^{-5}$ for SGD (2 epochs, larger steps diverge); $\gamma = 0.5$ and 1000 iterations for the logistic models. λ was selected on the validation F1 over $\{10^{-8}, \dots, 10^{-1}\}$.

Non-linear models. We also implemented a decision tree (Gini impurity, features binned into 32 quantile bins to make split search fast), a random forest (50 trees of depth 12, $\sqrt{D}$ candidate features per split, bootstrap samples), an ensemble averaging the probabilities of the logistic regression and the forest, $p = \alpha\, p_{\log} + (1 - \alpha)\, p_{\text{forest}}$, and gradient boosting in which small regression trees successively correct the logit of the logistic regression (logistic

Table I
Comparison of the models on the same validation split (best threshold for each model).

Model	F1	Accuracy
MSE with SGD	0.417	0.849
MSE with GD	0.423	0.859
Least squares	0.425	0.865
Ridge regression ($\lambda = 10^{-4}$)	0.425	0.865
Logistic regression	0.428	0.875
Reg. logistic regression ($\lambda = 10^{-4}$)	0.429	0.875
Decision tree (depth 8)	0.409	0.864
Random forest (50 trees)	0.426	0.865
Ensemble log. + forest ($\alpha = 0.4$)	0.431	0.881
Logistic + boosting	0.436	–
Neural network (64 hidden units)*	0.417	0.878

*Own feature set, measured on a different holdout split.

Table II
Effect of each preprocessing step (cumulative). Regularized logistic regression, validation F1 on the 80/20 split.

Preprocessing	Features	F1
Raw (empty/constant cols. removed)	218	0.409
+ cleaning of codes and columns	170	0.422
+ one-hot of 7 nominal variables	199	0.425
+ missing indicators (not kept)	293	0.425
+ interactions and squares	207	0.427

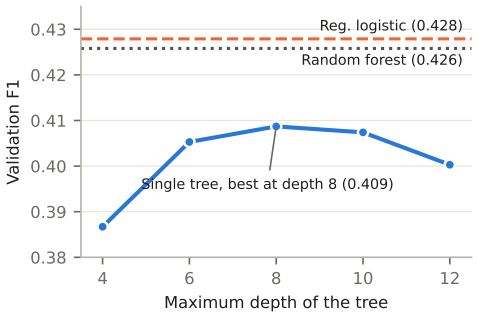


Figure 1. Validation F1 of a single decision tree versus its maximum depth.

loss, second-order split gain as in XGBoost [2], 100 rounds of depth-5 trees, learning rate 0.1, half of the rows and features per tree). Finally, a neural network with one hidden layer of 64 ReLU units (Adam [3], positive class weight 3, early stopping) was evaluated on its own feature set (one-hot sex, marital status and employment, missing-value indicators) and holdout split.

IV. Results

Preprocessing. Table II shows that cleaning the answer codes gives the largest gain (+0.013 F1): treating "refused" (9) as a large number misleads a linear model. One-hot encoding and the interactions each add a small improvement, while missing-value indicators add 94 columns without any gain, so they were not kept.

Linear models. Table I shows that the two logistic models beat the four regression models: the squared loss also penalizes confident correct predictions, unlike the logistic loss. Regularization barely matters: with $N \gg D$ the model does not overfit, and $\lambda \geq 10^{-2}$ only lowers the F1. However, $X^\top X$ is singular (smallest eigenvalue ≈ 0) because several computed BRFSS variables are linear combinations of others, so plain least squares produces very large weights and ridge regression is the numerically stable choice. SGD needs a tiny step because rare one-hot categories become very large after standardization.

Non-linear models. A single tree is clearly worse than the linear models (Fig. 1): shallow trees underfit and deeper trees overfit. Averaging 50 trees in a random forest almost closes the gap, and averaging the forest with the logistic regression gives a small gain (0.431). Boosting the logistic regression is the best model (0.436), confirmed by 5-fold CV (0.4314 ± 0.0040 vs. 0.4268 ± 0.0035 for the logistic regression): our hand-made products miss some interactions. The neural network did not beat the linear models (0.417).

Final model. We submitted the regularized logistic regression ($\lambda = 10^{-4}$), retrained on all the training data with the threshold chosen on the validation split ($t = 0.206$). Its 5-fold CV F1 is 0.4268 ± 0.0035 (accuracy 0.871 ± 0.007, precision 0.35, recall 0.54), and it obtained F1 = 0.441 and accuracy = 0.878 on AIcrowd, our best leaderboard score. It finds about half of the sick persons, with roughly two false alarms per true detection.

V. Discussion

We kept the model with our best leaderboard score. Boosting is the most promising direction (+0.005 in CV). All reasonable methods reach an F1 between 0.42 and 0.44, so the information in the survey seems to be the main limit. The label is a past diagnosis, so the model detects the disease rather than predicting it; strong predictors such as difficulty walking may even be consequences of it.

References

[1] Centers for Disease Control and Prevention, "Behavioral Risk Factor Surveillance System, 2015 survey data," https://www.cdc.gov/brfss/annual_data/annual_2015.html.

[2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. KDD, 2016, pp. 785–794.

[3] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. ICLR, 2015.